

[Vitamin D: Fact Sheet for Consumers \(original version\)](#).

Vitamin D: Fact Sheet for Consumers

Table of Contents

- [What is vitamin D and what does it do?](#)
- [How much vitamin D do I need?](#)
- [What foods provide vitamin D?](#)
- [Can I get vitamin D from the sun?](#)
- [What kinds of vitamin D dietary supplements are available?](#)
- [Am I getting enough vitamin D?](#)
- [What happens if I don't get enough vitamin D?](#)
- [What are some effects of vitamin D on health?](#)
- [Can vitamin D be harmful?](#)
- [Does vitamin D interact with medications or other dietary supplements?](#)
- [Vitamin D and healthful eating](#)
- [Where can I find out more about vitamin D?](#)
- [Disclaimer](#)
- [Glossary](#)

For information on vitamin D and COVID-19, see [Dietary Supplements in the Time of COVID-19](#).

What is vitamin D and what does it do?

Vitamin D is a nutrient you need for good health. It helps your body absorb calcium, one of the main building blocks for strong bones. Together with calcium, vitamin D helps protect you from developing osteoporosis, a disease that thins and weakens the bones and makes them more likely to break. Your body needs vitamin D for other functions too. Your muscles need it to move, and your nerves need it to carry messages between your brain and your body. Your immune system needs vitamin D to fight off invading bacteria and viruses.

How much vitamin D do I need?

The amount of vitamin D you need each day depends on your age. Average daily recommended amounts are listed below in micrograms (mcg) and International Units (IU).

Life Stage	Recommended Amount
Birth to 12 months	10 mcg (400 IU)
Children 1–13 years	15 mcg (600 IU)
Teens 14–18 years	15 mcg (600 IU)
Adults 19–70 years	15 mcg (600 IU)
Adults 71 years and older	20 mcg (800 IU)
Pregnant and breastfeeding teens and women	15 mcg (600 IU)

What foods provide vitamin D?

Very few foods naturally contain vitamin D. Fortified foods provide most of the vitamin D in the diets of people in the United States. Check the Nutrition Facts label for the amount of vitamin D in a food or beverage.

- Almost all of the U.S. milk supply is fortified with about 3 mcg (120 IU) vitamin D per cup. Many plant-based alternatives such as soy milk, almond milk, and oat milk are similarly fortified, but foods made from milk, like cheese and ice cream, are usually not fortified.
- Vitamin D is added to many breakfast cereals and to some brands of orange juice, yogurt, margarine, and other food products.
- Fatty fish (like trout, salmon, tuna, and mackerel) and fish liver oils are among the best natural sources of vitamin D.
- Beef liver, egg yolks, and cheese have small amounts of vitamin D.
- Mushrooms provide a little vitamin D. Some mushrooms have been exposed to ultraviolet light to increase their vitamin D content.

Can I get vitamin D from the sun?

Your body makes vitamin D when your bare skin is exposed to the sun. Most people get at least some vitamin D this way. However, clouds, smog, old age, and having dark-colored skin reduce the amount of vitamin D your skin makes. Also, your skin does not make vitamin D from sunlight through a window.

Ultraviolet radiation from sunshine can cause skin cancer, so it's important to limit how much time you spend in the sun. Although sunscreen limits vitamin D production, health experts recommend using sunscreen with a sun protection factor (SPF) of 15 or more when you're out in the sun for more than a few minutes.

What kinds of vitamin D dietary supplements are available?

Vitamin D is found in multivitamin/mineral supplements. It is also available in dietary supplements containing only vitamin D or vitamin D combined with a few other nutrients. The two forms of vitamin D in supplements are D2 (ergocalciferol) and D3 (cholecalciferol). Both forms increase vitamin D in your blood, but D3 might raise it higher and for longer than D2. Because vitamin D is fat soluble, it is best absorbed when taken with a meal or snack that includes some fat.

Am I getting enough vitamin D?

Because you get vitamin D from food, sunshine, and dietary supplements, one way to know if you're getting enough is a blood test that measures the amount of vitamin D in your blood. In the blood, a form of vitamin D known as 25-hydroxyvitamin D is measured in either nanomoles per liter (nmol/L) or nanograms per milliliter (ng/mL). One nmol/L is equal to 0.4 ng/mL. So, for example, 50 nmol/L is the same as 20 ng/mL.

- Levels of 50 nmol/L (20 ng/mL) or above are adequate for most people for bone and overall health.
- Levels below 30 nmol/L (12 ng/mL) are too low and might weaken your bones and affect your health.
- Levels above 125 nmol/L (50 ng/mL) are too high and might cause health problems.

In the United States, most people have adequate blood levels of vitamin D. However, almost one out of four people have vitamin D blood levels that are too low or inadequate for bone and overall health.

Some people are more likely than others to have trouble getting enough vitamin D:

- *Breastfed infants.* Breast milk alone does not provide infants with an adequate amount of vitamin D. Breastfed infants should be given a supplement of 10 mcg (400 IU) of vitamin D each day.
- *Older adults.* As you age, your skin's ability to make vitamin D when exposed to sunlight declines.
- *People who seldom expose their skin to sunshine* because they do not go outside or because they keep their body and head covered. Sunscreen also limits the amount of vitamin D your skin produces.
- *People with dark skin.* The darker your skin, the less vitamin D you make from sunlight exposure.
- *People with conditions that limit fat absorption*, such as Crohn's disease, celiac disease, or ulcerative colitis. This is because the vitamin D you consume is absorbed in the gut along with fat, so if your body has trouble absorbing fat, it will also have trouble absorbing vitamin D.
- *People with obesity or who have undergone gastric bypass surgery.* They may need more vitamin D than other people.

What happens if I don't get enough vitamin D?

In children, vitamin D deficiency causes rickets, a disease in which the bones become soft, weak, deformed, and painful. In teens and adults, vitamin D deficiency causes osteomalacia, a disorder that causes bone pain and muscle weakness.

What are some effects of vitamin D on health?

Scientists are studying vitamin D to better understand how it affects health. Here are several examples of what this research has shown.

Bone health and osteoporosis

Long-term shortages of vitamin D and calcium cause your bones to become fragile and break more easily. This condition is called osteoporosis. Millions of older women and men have osteoporosis or are at risk of developing this condition. Muscles are also important for healthy bones because they help maintain balance and prevent falls. A shortage of vitamin D may lead to weak, painful muscles.

Getting recommended amounts of vitamin D and calcium from foods (and supplements, if needed) will help maintain healthy bones and prevent osteoporosis. Taking vitamin D and calcium supplements slightly increases bone strength in older adults, but it's not clear whether they reduce the risk of falling or breaking a bone.

Cancer

Vitamin D does not seem to reduce the risk of developing cancer of the breast, colon, rectum, or lung. It is not clear whether vitamin D affects the risk of prostate cancer or chance of surviving this cancer. Very high blood levels of vitamin D may even increase the risk of pancreatic cancer.

Clinical trials suggest that while vitamin D supplements (with or without calcium) may not affect your risk of getting cancer, they might slightly reduce your risk of dying from this disease. More research is needed to better understand the role that vitamin D plays in cancer prevention and cancer-related death.

Heart disease

Vitamin D is important for a healthy heart and blood vessels and for normal blood pressure. Some studies show that vitamin D supplements might help reduce blood cholesterol levels and high blood pressure—two of the main risk factors for heart disease. Other studies show no benefits. If you are overweight or have obesity, taking vitamin D at doses above 20 mcg (800 IU) per day plus calcium might actually raise your blood pressure. Overall, clinical trials find that vitamin D supplements do not reduce the risk of developing heart disease or dying from it, even if you have low blood levels of the vitamin.

Depression

Vitamin D is needed for your brain to function properly. Some studies have found links between low blood levels of vitamin D and an increased risk of depression. However, clinical trials show that taking vitamin D supplements does not prevent or ease symptoms of depression.

Multiple sclerosis

People who live near the equator have more sun exposure and higher vitamin D levels. They also rarely develop multiple sclerosis (MS), a disease that affects the nerves that carry messages from the brain to the rest of the body. Many studies find a link between low blood vitamin D levels and the risk of developing MS. However, scientists have not actually studied whether vitamin D supplements can prevent MS. In people who

have MS, clinical trials show that taking vitamin D supplements does not keep symptoms from getting worse or coming back.

Type 2 diabetes

Vitamin D helps your body regulate blood sugar levels. However, clinical trials in people with and without diabetes show that supplemental vitamin D does not improve blood sugar levels, insulin resistance, or hemoglobin A1c levels (the average level of blood sugar over the past 3 months). Other studies show that vitamin D supplements don't stop most people with prediabetes from developing diabetes.

Weight loss

Taking vitamin D supplements or eating foods that are rich in vitamin D does not help you lose weight.

Can vitamin D be harmful?

Yes, getting too much vitamin D can be harmful. Very high levels of vitamin D in your blood (greater than 375 nmol/L or 150 ng/mL) can cause nausea, vomiting, muscle weakness, confusion, pain, loss of appetite, dehydration, excessive urination and thirst, and kidney stones. Extremely high levels of vitamin D can cause kidney failure, irregular heartbeat, and even death. High levels of vitamin D are almost always caused by consuming excessive amounts of vitamin D from dietary supplements. You cannot get too much vitamin D from sunshine because your skin limits the amount of vitamin D it makes.

The daily upper limits for vitamin D include intakes from all sources—food, beverages, and supplements—and are listed below in micrograms (mcg) and IU. However, your health care provider might recommend doses above these upper limits for a period of time to treat a vitamin D deficiency.

Ages	Upper Limit
Birth to 6 months	25 mcg (1,000 IU)
Infants 7–12 months	38 mcg (1,500 IU)
Children 1–3 years	63 mcg (2,500 IU)
Children 4–8 years	75 mcg (3,000 IU)
Children 9–18 years	100 mcg (4,000 IU)
Adults 19 years and older	100 mcg (4,000 IU)
Pregnant and breastfeeding teens and women	100 mcg (4,000 IU)

Does vitamin D interact with medications or other dietary supplements?

Yes, vitamin D supplements may interact with some medicines. Here are several examples:

- Orlistat (Xenical and alli) is a weight-loss drug. It can reduce the amount of vitamin D your body absorbs from food and supplements.
- Cholesterol-lowering statins might not work as well if you take high-dose vitamin D supplements. This includes atorvastatin (Lipitor), lovastatin (Altoprev and Mevacor), and simvastatin (FloLipid and Zocor)
- Steroids such as prednisone (Deltasone, Rayos, and Sterapred) can lower your blood levels of vitamin D.
- Thiazide diuretics (such as Hygroton, Lozol, and Microzide) could raise your blood calcium level too high if you take vitamin D supplements.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your

medicines. They can also explain whether the medicines you take might interfere with how your body absorbs or uses other nutrients.

Vitamin D and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible otherwise to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about vitamin D?

- For more information on vitamin D
 - Office of Dietary Supplements (ODS) [Health Professional Fact Sheet on Vitamin D](#)
 - MedlinePlus, [Vitamin D](#)
- For more information on food sources of vitamin D
 - ODS [Health Professional Fact Sheet on Vitamin D](#)
 - USDA's [FoodData Central](#)
 - Nutrient List for vitamin D (listed by [food](#) or by [vitamin D content](#)), USDA
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

bacteria

Single-celled organisms that are too small to be seen without a microscope. Bacteria are found everywhere and may be helpful or harmful.

blood sugar

The main source of energy used by the body's cells. Blood sugar comes from food and is made by the liver, and is carried to the cells through the bloodstream. Also called blood glucose.

blood vessel

A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

calcium

A mineral found throughout the body. Calcium is needed for healthy bones and teeth, for nerves and enzymes to function properly, and for blood clotting. Calcium is found in some foods, including milk, yogurt, and cheese, and in Chinese cabbage, kale, broccoli and fortified foods, such as many drinks, tofu, and cereals.

celiac disease

An autoimmune disorder in which eating gluten (a protein found in wheat, rye, barley, and possibly oats) causes the immune system to damage the small intestine, making it unable to absorb nutrients. It is a genetic disease that sometimes becomes active for the first time after surgery, pregnancy, childbirth, viral infection, or extreme stress. Also called sprue.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals

contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

clinical trial

A type of research study that uses volunteers to test the safety and efficacy (the ability to produce a beneficial effect) of new methods of screening (checking for disease when there are no symptoms), prevention, diagnosis, or treatment of a disease. Also called a clinical study.

colon

A tube-like organ about 5 feet long in adults that is connected to the small intestine at one end and the anus at the other. The colon absorbs water, some nutrients, and electrolytes (such as sodium and chloride) from partially digested food. The remaining material (solid waste called stool) moves through the colon to the rectum and leaves the body through the anus as a bowel movement. The colon is part of the digestive system (a series of organs from the mouth to the anus). Also called the large intestine.

consume

To eat or drink.

Crohn's disease

A long-lasting (chronic) disease that causes severe irritation in the gastrointestinal tract. It usually affects the lower small intestine (called the ileum) or the colon, but it can affect any part of the digestive tract from the mouth to the anus. It is painful, causing severe watery or bloody diarrhea, and may lead to life-threatening complications. Crohn's disease is a form of inflammatory bowel disease.

deficiency

An amount that is not enough; a shortage.

diabetes

A disease in which blood sugar (glucose) levels are high because the body is unable to use glucose properly. Diabetes occurs when the body does not make enough insulin, which helps the cells use glucose, or when the body no longer responds to insulin.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

fragile

Easily broken.

gastric

Having to do with the stomach.

hemoglobin

The substance inside red blood cells that carries oxygen from the lungs to the body's tissues.

high blood pressure

A blood pressure measurement of 140/90 mmHg (millimeters of mercury) or higher is considered high blood pressure (hypertension). Blood pressure is the force of blood pushing against the walls of the arteries. Blood pressure measurements are written as two numbers, for example 120/80. The first number (the systolic pressure) measures the pressure when the heart beats and pumps out blood into the arteries. The second number (the diastolic pressure) measures the pressure when the heart is at rest between beats. High blood pressure is a condition that occurs when a person's blood pressure often measures above 140/90 or regularly stays at that level or higher. This condition usually has no symptoms but can be life-threatening. It damages the arteries and increases the chance of stroke, heart attack, kidney failure, and blindness. Also called hypertension.

immune system

A group of organs and cells that defends the body against infection, disease, and altered (mutated) cells. It includes the thymus, spleen, lymphatic system (lymph nodes and lymph vessels), bone marrow, tonsils, and white blood cells.

infant

A child younger than 12 months old.

insulin resistance

A condition in which glucose (blood sugar) cannot be absorbed by the cells and used for energy. Instead, glucose builds up in the blood and the body produces more and more insulin (which normally would help glucose get into the cells), resulting in abnormally high blood levels of both glucose and insulin. This can lead to pre-diabetes, type 2 diabetes, and other serious health problems.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

International Unit

IU. A measurement used to measure the activity of some vitamins and other biological substances (such as enzymes and hormones).

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kidney failure

Loss of kidney function. When kidney failure is caused by an acute (sudden and short-term) condition, such as a traumatic injury or poisoning, the kidneys may be able to recover. Usually, however, kidney failure is caused by chronic (long-term) conditions such as high blood pressure and diabetes. Chronic diseases permanently damage kidney tissue; treatments include dialysis and kidney transplantation.

kidney stone

A "pebble" that forms in a kidney from salts and minerals in the urine. A small kidney stone is able to pass out of the body; a large stone may block the urinary tract and require medical help.

label

When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

liver

A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

lung

An organ in the chest that supplies oxygen to the body and removes carbon dioxide. There are two lungs in the body.

microgram

μg or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nerve

A bundle of microscopic fibers that carries messages back and forth from the brain to other parts of the body.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

nutrition

The process of eating, digesting, and absorbing nutrients (such as protein, carbohydrate, fat, vitamins, minerals, and water) from food to maintain the body, grow new cells, repair tissues, and supply energy. Nutrition is also the science of food, diet, and health.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

osteomalacia

A condition in adults in which bones become soft and deformed because they don't have enough calcium and phosphorus. It is usually caused by not having enough vitamin D in the diet, not getting enough sunlight, or a problem with the way the body uses vitamin D. Symptoms include bone pain and muscle weakness. When the condition occurs in children, it is called rickets.

osteoporosis

A condition in which bones become weak and brittle, increasing the chance they may break.

overweight

An excess amount of body weight that includes muscle, bone, fat, and water. Overweight can be assessed by calculating the body mass index (BMI). (BMI is a number that estimates the amount of body fat on a person, based on weight and height. An adult with a BMI between 25 and 29.9 is considered overweight. Some people, such as bodybuilders or other athletes with a lot of muscle, can be overweight without having obesity. See: obesity.

pancreatic cancer

Uncontrolled growth of abnormal cells in the pancreas, an organ behind the stomach that makes pancreatic juices to help digest food, and several hormones, including insulin.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

prevention

In medicine, action taken to decrease the chance (risk) of developing a disease.

prostate cancer

Uncontrolled growth of abnormal cells in the prostate (a gland in the male reproductive system found below the bladder and in front of the rectum). Prostate cancer usually occurs in older men.

regulate

To govern, make uniform, and bring under the control of a rule, principle, or legal system. In the United States, the FDA has the authority to regulate dietary supplements.

rickets

A condition in children in which bones become soft and deformed because they don't have enough calcium and phosphorus. It is caused by not having enough vitamin D in the diet or by not getting enough sunlight. In adults, this condition is called osteomalacia.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

risk factor

Something that may increase the chance of developing a disease. For example, a diet that is low in calcium and vitamin D is a risk factor for osteoporosis; smoking is a risk factor for lung cancer.

skin cancer

Cancer that forms in tissues of the skin. Most skin cancers form in older people on parts of the body exposed to the sun or in people who have weakened immune systems.

soy

A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

SPF

Sun protection factor. A scale for rating the level of sunburn protection in sunscreen products. The higher the SPF, the more sunburn protection it gives.

statin

A drug used to treat high cholesterol. Statins lower the amount of cholesterol and certain fats in the blood.

sunscreen

A substance that helps protect the skin from the sun's harmful rays. Sunscreens reflect, absorb, and scatter ultraviolet A and ultraviolet B light to provide protection against both types of light. Using lotions, creams, or gels that contain sunscreens can help protect the skin from premature aging and damage that may lead to skin cancer.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

thiazide diuretic

A drug used in the treatment of high blood pressure and swelling caused by excess fluid in body tissues (edema). It increases the amount of urine made by the body.

ulcerative colitis

Chronic inflammation of the colon that causes ulcers to form in its lining. This condition is marked by abdominal pain, cramps, and loose discharges of pus, blood, and mucus from the bowel.

ultraviolet light

Invisible rays that are part of the energy that comes from the sun. Ultraviolet light also comes from sun lamps and tanning beds. It can damage the skin and cause skin cancer.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

virus

An organism that can grow and multiply only inside the cells of living humans, plants, or animals. It is able to change (mutate) as it multiplies, which makes viral illnesses difficult to treat. Viruses cause many infections and diseases such as the common cold, AIDS (acquired immunodeficiency syndrome), herpes, and hepatitis.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin D

A nutrient that is obtained from the diet and can be made in the skin after exposure to sunlight. Vitamin D acts as a hormone. It helps to form and maintain strong bones, maintain normal blood levels of calcium and phosphorus, and increase calcium absorption; it also helps to maintain a healthy immune system and control cell growth. Vitamin D is found in some foods, including some types of fatty fish, and milk and breakfast cereals that are fortified with vitamin D.

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